



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

AI Based Intrusion Detection System (AI IDS)

The domain of the Project:
AI & Network Security

Team Mentor :Mr. Derick Johnson

Team Members:
Ms. Venkata Naga Sahithya Mitta

Period of the project: 2 Months

February 2026 to August 2026



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Declaration

The project titled AI Based Intrusion Detection System (IDS) has been mentored by Mr. Derick Jhonson organised by SURE Trust, from February 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Ms. Sahithya

Mentor's Name

Mr. Derick Jhonson

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

The rapid growth of internet usage, cloud computing, Internet of Things (IoT) devices, and digital communication has significantly increased the number and complexity of cyber attacks. Organizations continuously face security threats such as denial-of-service attacks, unauthorized access, malware infections, phishing campaigns, and network intrusions. Traditional security systems primarily rely on signature-based detection techniques, which are effective only against previously known attacks and often fail to identify new or sophisticated threats. This limitation creates a need for intelligent, automated, and adaptive security solutions capable of detecting suspicious network activities in real time.

The **AI Intrusion Detection System (AI IDS)** was developed to address these challenges by integrating Artificial Intelligence and Machine Learning into network security monitoring. The system is designed to analyze network traffic continuously, classify packets as either **Normal** or **Attack**, and provide security analysts with real-time visibility into potential cyber threats through an interactive Security Operations Center (SOC) dashboard. By automating the intrusion detection process, the project reduces manual effort, improves response time, and enhances the overall efficiency of security monitoring.

The proposed system follows a modular architecture consisting of a machine learning engine, a Flask-based backend, RESTful APIs, and a web-based frontend dashboard. The machine learning model processes network traffic features and predicts whether the observed traffic is benign or malicious. Based on the prediction confidence, the system automatically assigns severity levels such as Low, Medium, High, or Critical, enabling analysts to prioritize incidents according to their potential impact. The backend manages data processing, alert generation, report creation, and communication between different system components, while the frontend provides a centralized interface for monitoring network activities and security events.

The dashboard includes several functional modules that improve the overall usability of the system. These modules provide real-time packet monitoring, live security alerts, traffic analytics, user profile management, configurable severity rules, threat source analysis, security report generation, system settings, and notification management. The system also supports exporting security reports in CSV and PDF formats, making it suitable for documentation, auditing, and incident analysis. Additionally, email and Telegram notification features can be configured to deliver critical security alerts immediately to administrators, enabling faster incident response.

During the implementation phase, Python was used as the primary programming language, with the Flask framework handling backend services and REST API communication. Machine



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learning algorithms available through Scikit-learn were utilized for network traffic classification, while Pandas and NumPy were used for data preprocessing and analysis. The frontend was developed using modern web technologies to provide an intuitive and responsive user interface capable of presenting live security information in an organized manner.

The developed AI IDS successfully demonstrates the practical application of Artificial Intelligence in cybersecurity. It provides centralized monitoring, intelligent intrusion detection, automated alert management, and comprehensive reporting within a single platform. The modular design allows the system to be easily extended with additional features such as deep learning-based threat detection, real-time packet capture from enterprise networks, cloud deployment, SIEM integration, threat intelligence feeds, automated incident response, and predictive cyber threat analytics.

In conclusion, the AI Intrusion Detection System represents an efficient, scalable, and user-friendly cybersecurity solution that enhances network visibility and supports security professionals in detecting, analyzing, and responding to cyber threats more effectively. The project highlights the growing importance of Artificial Intelligence in modern cybersecurity and demonstrates how intelligent automation can strengthen organizational security while reducing operational complexity and improving decision-making for Security Operations Center (SOC) analysts.



Introduction

The rapid advancement of digital technologies, cloud computing, Internet of Things (IoT), and online communication has transformed the way organizations operate and exchange information. While these technologies provide significant benefits, they have also increased the risk of cyber attacks targeting networks, systems, and sensitive data. Modern organizations experience thousands of network events every day, making it difficult for security analysts to manually monitor all activities and identify malicious behavior in a timely manner. As cyber threats continue to evolve in complexity and frequency, there is a growing need for intelligent security solutions that can automatically detect, analyze, and respond to suspicious network activities.

An Intrusion Detection System (IDS) is one of the essential components of modern cybersecurity. Its primary purpose is to monitor network traffic, identify unauthorized or malicious activities, and alert security administrators before an attack causes significant damage. Traditional IDS solutions mainly depend on signature-based detection methods, which compare network traffic against predefined attack signatures. Although these systems are effective in detecting known attacks, they often fail to identify new or unknown threats such as zero-day attacks, advanced persistent threats (APTs), and evolving malware. In addition, managing large volumes of security logs manually is both time-consuming and error-prone.

To overcome these challenges, Artificial Intelligence (AI) and Machine Learning (ML) have become increasingly important in cybersecurity. Machine learning algorithms can learn traffic patterns from historical data, recognize abnormal behavior, and classify network packets with high accuracy. Unlike traditional methods, ML-based intrusion detection systems can identify both known and previously unseen attack patterns, making them more effective in dynamic network environments. This capability enables faster threat detection, reduces false alarms, and supports proactive security monitoring.

The **AI Intrusion Detection System (AI IDS)** developed in this project combines Machine Learning with a web-based Security Operations Center (SOC) dashboard to provide intelligent and real-time network monitoring. The system analyzes network traffic, classifies packets as **Normal** or **Attack**, assigns confidence scores and severity levels, and presents the results through an interactive dashboard. The dashboard includes modules for traffic analytics, live security alerts, threat source analysis, report generation, user profile management, configurable severity rules, notification services, and system settings. These features help security analysts monitor the overall security posture of the network and respond quickly to potential threats.



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The backend of the system is implemented using **Python** and the **Flask** framework, which manages data processing, machine learning predictions, alert generation, and REST API communication. The frontend provides a responsive and user-friendly interface that refreshes continuously to display the latest security information. The modular architecture adopted in this project improves maintainability, scalability, and future enhancement of the system.

This project demonstrates the practical application of Artificial Intelligence in network security by automating intrusion detection and reducing the dependency on manual analysis. It provides an efficient, scalable, and cost-effective solution for monitoring network traffic, identifying cyber threats, and supporting informed decision-making within a Security Operations Center (SOC). The proposed system can be further extended with advanced machine learning models, cloud deployment, real-time packet capture, SIEM integration, and automated incident response, making it suitable for both academic research and real-world cybersecurity applications.



Project Objectives

Clearly Defined Objectives and Goals of the Project

The primary objective of the **AI Intrusion Detection System (AI IDS)** is to develop an intelligent and automated cybersecurity solution capable of monitoring network traffic, detecting malicious activities, and assisting security analysts in identifying potential threats in real time. The project integrates Machine Learning with a web-based dashboard to improve the efficiency, accuracy, and speed of intrusion detection while reducing manual monitoring efforts.

The specific objectives of this project are:

1. To design and develop a web-based AI Intrusion Detection System for continuous network security monitoring.
2. To implement a Machine Learning model capable of classifying network traffic as **Normal** or **Attack** based on traffic features.
3. To provide a real-time Security Operations Center (SOC) dashboard for monitoring network events, packet statistics, and threat levels.
4. To generate live security alerts with prediction confidence, severity level, source IP address, destination IP address, and attack information.
5. To analyze network traffic through graphical visualizations, traffic analytics, and threat score calculations for better decision-making.
6. To identify high-risk source IP addresses and targeted destination systems through the Threat Sources module.
7. To provide configurable severity rules that automatically classify detected attacks into Low, Medium, High, and Critical categories based on prediction confidence.
8. To generate security reports in CSV and PDF formats for auditing, incident analysis, and documentation purposes.
9. To support Email and Telegram notifications for critical security incidents, enabling faster response by security administrators.
10. To develop a modular, scalable, and user-friendly cybersecurity platform that can be extended with advanced Machine Learning models, real-time packet capture, SIEM integration, cloud deployment, and automated incident response mechanisms.



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Expected Outcomes and Deliverables

The expected outcome of this project is the successful development of a fully functional AI-based Intrusion Detection System capable of detecting suspicious network activities and presenting the results through an interactive web dashboard. The system provides centralized monitoring, real-time alert generation, traffic analytics, threat source analysis, severity management, and automated security reporting. It also delivers notification support through Email and Telegram for critical incidents.

The final deliverables of the project include a trained Machine Learning intrusion detection model, a Flask-based backend application, REST APIs for data communication, a responsive dashboard with multiple security modules, report generation functionality, notification services, project documentation, and the complete source code repository. These deliverables demonstrate the practical application of Artificial Intelligence in cybersecurity and provide a foundation for future enhancements in intelligent network defense systems.



Methodology and Results

The AI Intrusion Detection System (AI IDS) was developed using a Machine Learning-based approach to monitor network traffic and identify malicious activities in real time. The system follows a layered architecture that integrates data processing, machine learning, backend services, and a web-based dashboard to provide an intelligent Security Operations Center (SOC) solution.

Initially, network traffic data is collected from the intrusion detection log files. The collected data contains various network attributes such as source IP address, destination IP address, protocol information, packet characteristics, timestamps, and traffic features. Before applying machine learning, the data is preprocessed by cleaning invalid entries, selecting useful features, and transforming the data into a format suitable for prediction.

The processed data is then passed to a trained Machine Learning model that classifies each network packet as either **Normal** or **Attack**. Along with the prediction, the model generates a confidence score, which is further used to assign severity levels such as Low, Medium, High, and Critical. This automated classification enables security analysts to identify high-risk activities more efficiently.

The backend of the application is developed using the Flask framework. It manages the interaction between the Machine Learning model and the frontend dashboard by exposing REST APIs. These APIs process incoming requests, retrieve prediction results, maintain IDS logs, calculate dashboard statistics, generate reports, and provide configuration settings. The backend also supports Email and Telegram notification services for critical security alerts.

The frontend provides an interactive Security Operations Center dashboard where users can monitor network traffic, review security alerts, analyze attack trends, generate reports, manage severity rules, view threat sources, and configure notification settings. The dashboard automatically refreshes at regular intervals to display the latest intrusion detection results.

The overall methodology combines Machine Learning, real-time monitoring, REST API communication, and interactive visualization to create an intelligent, scalable, and user-friendly network intrusion detection system capable of supporting cybersecurity analysts in identifying and responding to potential threats.



4.2 Software Used

The development of the AI Intrusion Detection System (AI IDS) involved various software tools, programming frameworks, libraries, and development environments. These tools were selected to support machine learning-based threat detection, backend development, frontend visualization, data processing, and system testing.

Programming Language

Python

Python was used as the primary programming language for implementing the backend logic, data processing pipeline, and Machine Learning model integration. Its extensive ecosystem of cybersecurity and data science libraries made it suitable for developing an intelligent intrusion detection system.

Backend Framework

Flask

Flask was used to develop the backend services and REST APIs of the AI IDS application. It handles communication between the Machine Learning model, database/log files, and the frontend dashboard. Flask APIs are responsible for providing security statistics, alert information, traffic analytics, threat analysis, and report generation.

Machine Learning Libraries

Scikit-learn

Scikit-learn was used for implementing and managing Machine Learning algorithms for network traffic classification. The library provides efficient tools for model training, testing, prediction, and performance evaluation.

Pandas

Pandas was used for data preprocessing, analysis, and handling IDS log data. It helps in cleaning, organizing, and transforming network traffic records before providing them to the Machine Learning model.

NumPy

NumPy was used for numerical operations and efficient handling of structured data during the preprocessing and prediction stages.

Frontend Technologies

HTML, CSS, and JavaScript



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These technologies were used to design and develop the interactive web dashboard interface. They provide the structure, styling, and dynamic behavior required for displaying real-time security information.

Chart.js

Chart.js was used for creating graphical visualizations such as traffic distribution, attack trends, threat scores, and security analytics. These visualizations help analysts understand network behaviour quickly.

Database and Storage

JSON File Storage

JSON-based storage was used for maintaining user information, configuration settings, and IDS-related records. This lightweight storage approach allows efficient data retrieval and management during system operation.

Development Environment

Visual Studio Code

Visual Studio Code was used as the primary development environment for writing, debugging, and managing project files.

Git and GitHub

Git was used for source code management and version control, while GitHub was used for maintaining the project repository and tracking development progress.

Testing and Monitoring Tools

Web Browser

A web browser was used to access and test the AI IDS dashboard interface, verify API responses, and validate real-time monitoring functionality.

The combination of these tools enabled the successful development of an AI-powered intrusion detection platform with machine learning-based classification, automated alert generation, and real-time security visualization capabilities.

4.3 Data Collection Approach

The AI Intrusion Detection System (AI IDS) requires network traffic data to identify and classify malicious activities. The data collection process plays an important role in training the Machine Learning model and evaluating the performance of the intrusion detection system.

The proposed system uses network traffic records collected from intrusion detection datasets and generated IDS log files. These records contain detailed information about



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network communication activities, including source and destination details, protocol information, traffic behaviour, and attack characteristics.

Dataset Source

The Machine Learning model was trained and evaluated using the NSL-KDD dataset, which is a widely used benchmark dataset in the field of network intrusion detection. The dataset contains labeled network traffic records categorized into normal and attack classes.

The dataset consists of multiple network features representing different characteristics of network connections, such as:

- Duration of network connection
- Protocol type
- Service information
- Source bytes
- Destination bytes
- Connection status
- Error rates
- Traffic behavior patterns
- Network activity statistics

Each network record is assigned a class label that identifies whether the traffic is:

- Normal Traffic
- Attack Traffic

This labeled information enables the Machine Learning model to learn patterns associated with malicious network activities.

Data Preprocessing

Before applying the Machine Learning model, the collected network data undergoes multiple preprocessing steps to improve detection accuracy and model performance.

The preprocessing steps include:

Data Cleaning:

The dataset is analyzed to identify missing values, duplicate records, and inconsistent entries. Invalid data records are removed to maintain data quality.



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Feature Selection:

Important network features are selected from the dataset to reduce complexity and improve the efficiency of the Machine Learning model.

Data Transformation:

Categorical attributes such as protocol type and service information are converted into numerical values using encoding techniques so that they can be processed by Machine Learning algorithms.

Data Splitting:

The processed dataset is divided into training and testing datasets.

- Training Data: Used for learning attack patterns.
- Testing Data: Used for evaluating model performance.

Machine Learning Model Input Flow

The processed network traffic records are provided as input to the trained Machine Learning model. The model analyzes traffic patterns and predicts whether the activity is normal or malicious.

The prediction workflow follows:

Network Traffic Data



Data Preprocessing



Feature Extraction



Machine Learning Model



Attack Classification



Alert Generation



Dashboard Visualization



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The model generates the following outputs:

- Prediction Result (Normal/Attack)
- Prediction Confidence Score
- Severity Level
- Security Alert Information

IDS Log Data Processing

After prediction, the generated security results are stored in IDS log files. These logs maintain important information required for security monitoring.

The IDS logs contain:

- Timestamp
- Source IP Address
- Destination IP Address
- Protocol Information
- Prediction Result
- Confidence Score
- Severity Level
- Alert Status

The backend application continuously processes these logs and provides updated security information to the dashboard through REST APIs.

Real-Time Data Monitoring Approach

The system follows a continuous monitoring approach where incoming network events are analyzed automatically by the AI model. The dashboard refreshes periodically and displays updated security information including alerts, traffic statistics, threat scores, and attack analysis.

This data collection and processing approach enables the AI IDS platform to provide accurate threat detection, automated classification, and efficient security monitoring capabilities.



4.4 Project Architecture

The AI Intrusion Detection System (AI IDS) follows a layered architecture that integrates Machine Learning, backend services, data storage, and a web-based Security Operations Center (SOC) dashboard. The architecture is designed to provide automated network monitoring, intelligent threat detection, alert generation, and security analysis.

The overall system architecture consists of the following major layers:

1. Data Collection Layer

The Data Collection Layer is responsible for gathering network traffic information required for intrusion detection. The system collects network event data from IDS logs and datasets containing network communication details.

The collected data includes:

- Source IP Address
- Destination IP Address
- Protocol Information
- Packet Details
- Network Traffic Features
- Timestamp Information

The collected traffic data is forwarded to the preprocessing module for further analysis.

2. Data Processing and Feature Engineering Layer

The Data Processing Layer prepares the collected network data before providing it to the Machine Learning model.

The major functions of this layer include:

- Cleaning raw network data
- Removing invalid records
- Handling missing values
- Selecting important network features
- Converting categorical values into numerical format
- Preparing model-ready input data



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This layer improves the quality of input data and helps the Machine Learning model achieve better detection performance.

3. Machine Learning Detection Layer

The Machine Learning Layer is the core component of the AI IDS system. It is responsible for analyzing network traffic patterns and identifying potential cyber threats.

The trained Machine Learning model classifies network activities into:

- Normal Traffic
- Attack Traffic

The model also generates:

- Prediction Confidence Score
- Attack Classification Result
- Risk Level

The Gradient Boosting Algorithm Machine Learning algorithm was selected as the final detection model due to its high accuracy and ability to identify complex attack patterns.

4. Backend Application Layer

The Backend Layer acts as a communication bridge between the Machine Learning model, database/log storage, and the frontend dashboard.

The backend application is developed using the Flask framework and provides REST API services.

The backend performs the following operations:

- Receives network analysis requests
- Processes IDS logs
- Communicates with Machine Learning model
- Calculates threat statistics
- Generates security reports
- Provides dashboard data through APIs
- Manages user information and system settings

The backend also supports automated notification services for critical security incidents.



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5. Security Operations Center (SOC) Dashboard Layer

The Frontend Dashboard Layer provides an interactive interface for security analysts to monitor and analyze network activities.

The dashboard contains multiple security modules:

Dashboard Module:

Displays overall system status, threat level, packet statistics, model confidence, and security overview.

Alerts Module:

Shows detected security events with information such as source IP, destination IP, prediction result, severity, and confidence score.

Traffic Analytics Module:

Provides graphical analysis of normal and malicious traffic patterns.

Profile Module:

Manages user account details and analyst information.

Severity Rules Module:

Allows configuration of alert classification levels such as Low, Medium, High, and Critical.

Reports Module:

Generates security reports in CSV and PDF formats for auditing and incident analysis.

Threat Sources Module:

Identifies high-risk source IP addresses and targeted destinations.

Settings Module:

Provides configuration options for thresholds, notifications, and automated response features.

Notifications Module:

Supports Email and Telegram alerts for critical security events.

6. Notification and Alert Response Layer

The Notification Layer improves incident response by sending alerts when critical security events are detected.

The system supports:

- Email-based security notifications



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- Telegram-based instant alerts
- Critical threat warnings

This allows security teams to respond quickly to high-risk incidents.

System Data Flow

The complete workflow of the AI IDS system is:

Network Traffic Data



Data Collection



Data Preprocessing



Feature Extraction



Machine Learning Classification



Attack Detection



Severity Calculation



Alert Generation



SOC Dashboard Visualization



Report Generation and Notifications



4.5 Backend Architecture

The backend architecture of the AI Intrusion Detection System (AI IDS) is responsible for managing communication between the Machine Learning model, data storage, security logs, notification services, and the frontend dashboard. The backend acts as the core processing layer that handles data analysis, API communication, and security event management.

The backend application is developed using the Flask framework because of its lightweight structure, flexibility, and ability to create RESTful APIs for real-time communication between different system components.

Backend Components

1. Flask Application Server

The Flask server provides the main execution environment for the AI IDS backend. It manages incoming requests from the frontend dashboard and processes security-related operations.

The Flask application performs:

- Running the AI IDS backend services
- Handling API requests and responses
- Managing dashboard data flow
- Connecting Machine Learning predictions with user interface
- Processing security events

2. REST API Layer

```
127.0.0.1:5000/api/alerts
Pretty print
{
  "alerts": [
    {
      "acknowledged": false,
      "alert_id": "e2b38c2ffebc6c",
      "assigned_to": "",
      "byte_count": 66,
      "confidence": 0.7926,
      "dst_ip": "35.223.238.178",
      "dst_port": 443,
      "flow_duration": 0.0,
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      "packet_count": 1,
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      "protocol": 6,
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      "severity": "Medium",
      "src_ip": "18.41.189.232",
      "src_port": 59777,
      "state_updated_at": "",
      "timestamp": "2026-08-03 19:24:24"
    },
    {
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      "alert_id": "484b8d97c9b6d2e",
      "assigned_to": "",
      "byte_count": 181,
      "confidence": 0.8438,
      "dst_ip": "18.41.189.232",
      "dst_port": 51888,
      "flow_duration": 0.0,
      "incident_status": "Open",
      "packet_count": 1,
      "prediction": "attack",
      "protocol": 17,
      "remediation": "Review alert and investigate source/destination.",
      "severity": "Medium",
      "src_ip": "18.41.189.242",
      "src_port": 53,
      "state_updated_at": "",
      "timestamp": "2026-08-03 19:24:15"
    },
    {
      "acknowledged": false,
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      "assigned_to": "",
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      "dst_ip": "18.41.189.232",
      "dst_port": 62825,
      "flow_duration": 0.0,
      "incident_status": "Open",
      "packet_count": 1,
      "prediction": "attack",
      "protocol": 17,
      "remediation": "Review alert and investigate source/destination.",
      "severity": "Medium",
      "src_ip": "18.41.189.242",
      "src_port": 53,
      "state_updated_at": "",
      "timestamp": "2026-08-03 19:24:15"
    }
  ]
}
```



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```
127.0.0.1:5000/api/stats
Gmail YouTube Maps
Pretty print
{
  "attack_packets": 1395,
  "attack_rate": 98.6,
  "attacks": 1395,
  "average_confidence": 85.2,
  "critical_alerts": 298,
  "high_alerts": 122,
  "last_event": "2026-08-03 19:24:33",
  "latest_event": "2026-08-03 19:24:33",
  "low_alerts": 20,
  "medium_alerts": 975,
  "normal": 20,
  "normal_packets": 20,
  "normal_rate": 1.4,
  "recent_attacks": 100,
  "source_scores": [
    {
      "attack_rate": 99.7,
      "attacks": 780,
      "avg_confidence": 84.8,
      "ip": "10.41.100.212",
      "level": "critical",
      "score": 100,
      "total": 780
    },
    {
      "attack_rate": 96.4,
      "attacks": 481,
      "avg_confidence": 83.9,
      "ip": "10.41.100.242",
      "level": "critical",
      "score": 100,
      "total": 499
    },
    {
      "attack_rate": 100.0,
      "attacks": 12,
      "avg_confidence": 91.2,
      "ip": "172.217.115.4",
      "level": "critical",
      "score": 100,
      "total": 12
    },
    {
      "attack_rate": 100.0,
      "attacks": 5,
      "avg_confidence": 91.9,
      "ip": "172.217.117.4",
      "level": "critical",
      "score": 100,
      "total": 5
    },
    {
      "attack_rate": 100.0,
      "attacks": 10,
      "avg_confidence": 85.5,
      "ip": "172.217.119.4"
    }
  ]
}
```

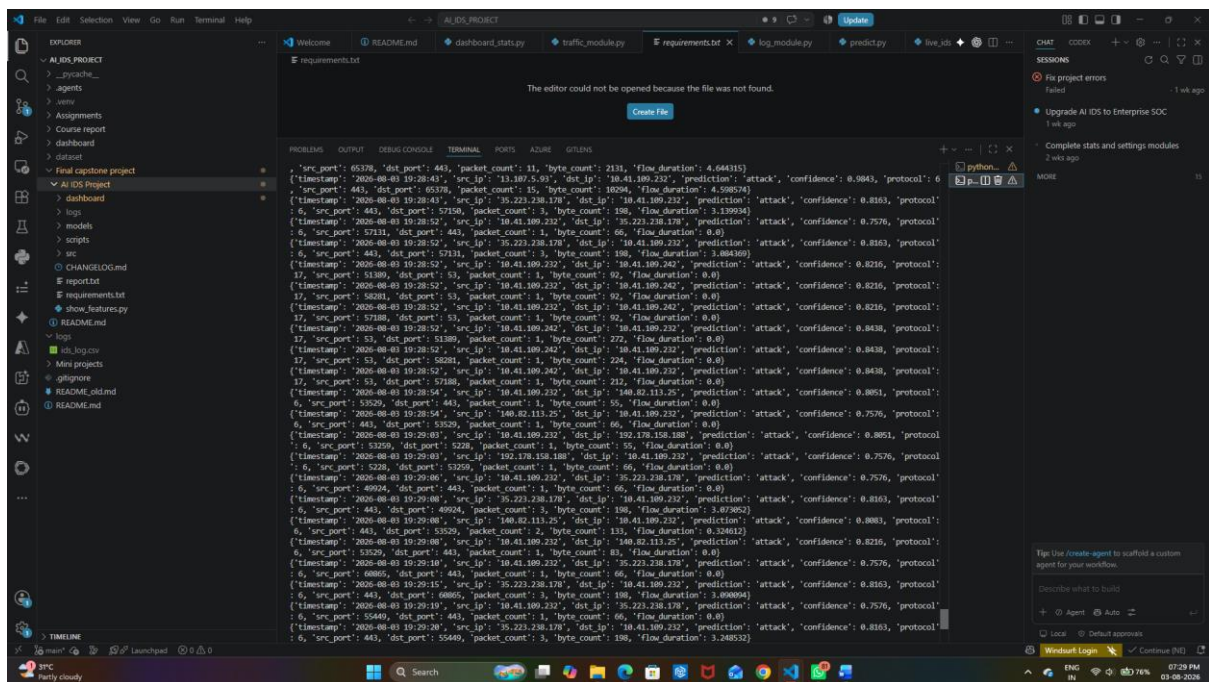
```
127.0.0.1:5000/api/traffic
Gmail YouTube Maps
Pretty print
{
  "attack_events": 1502,
  "attack_rate": 98.58,
  "avg_bytes_per_flow": 477.69,
  "avg_packets_per_flow": 3.11,
  "field_coverage": {
    "flow_volume": true,
    "ports": true,
    "protocol": true
  },
  "hourly_heatmap": [
    {
      "attacks": 1502,
      "hour": 19,
      "normal": 22,
      "total": 1524
    }
  ],
  "normal_events": 22,
  "protocol_distribution": [
    {
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      "protocol": "udp"
    },
    {
      "count": 680,
      "protocol": "tcp"
    },
    {
      "count": 1,
      "protocol": "3"
    }
  ],
  "severity_distribution": {
    "critical": 801,
    "high": 126,
    "low": 22,
    "medium": 1075
  },
  "timeline": [
    {
      "attacks": 150,
      "normal": 1,
      "time": "19:07",
      "total": 151
    },
    {
      "attacks": 109,
      "normal": 4,
      "time": "19:08",
      "total": 113
    },
    {
      "attacks": 155,
      "normal": 4,
      "time": "19:09",
      "total": 159
    }
  ]
}
```



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```
127.0.0.1:5000/api/severity
Gmail YouTube Maps
Pretty print
{
  "counts": {
    "Critical": 349,
    "High": 131,
    "Low": 23,
    "Medium": 1187
  },
  "recent": [
    {
      "byte_count": 418,
      "confidence": 0.791,
      "dst_ip": "18.41.189.232",
      "dst_port": 62825,
      "flow_duration": 14.987743,
      "packet_count": 3,
      "prediction": "attack",
      "protocol": 17,
      "remediation": "Review alert and investigate source/destination.",
      "severity": "Medium",
      "src_ip": "18.41.189.242",
      "src_port": 53,
      "timestamp": "2026-08-03 19:26:04"
    },
    {
      "byte_count": 264,
      "confidence": 0.8527,
      "dst_ip": "18.41.189.242",
      "dst_port": 53,
      "flow_duration": 14.995998,
      "packet_count": 3,
      "prediction": "attack",
      "protocol": 17,
      "remediation": "Review alert and investigate source/destination.",
      "severity": "High",
      "src_ip": "18.41.189.232",
      "src_port": 62825,
      "timestamp": "2026-08-03 19:26:04"
    },
    {
      "byte_count": 407,
      "confidence": 0.9728,
      "dst_ip": "18.41.189.232",
      "dst_port": 62864,
      "flow_duration": 0.147762,
      "packet_count": 6,
      "prediction": "attack",
      "protocol": 8,
      "remediation": "Review alert and investigate source/destination.",
      "severity": "Critical",
      "src_ip": "18.41.189.242",
      "src_port": 53,
      "timestamp": "2026-08-03 19:26:04"
    },
    {
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      "dst_ip": "18.41.189.232",
      "dst_port": 62784,
      "flow_duration": 0.147762,
      "packet_count": 6,
      "prediction": "attack",
      "protocol": 8,
      "remediation": "Review alert and investigate source/destination.",
      "severity": "Critical",
      "src_ip": "18.41.189.242",
      "src_port": 53,
      "timestamp": "2026-08-03 19:26:04"
    }
  ]
}
```

```
127.0.0.1:5000/api/reports/summary
Gmail YouTube Maps
Pretty print
{
  "attacks": 1859,
  "critical": 393,
  "csv_url": "//reports/export/csv",
  "normal": 23,
  "pdf_url": "//reports/export/pdf",
  "total": 1882
}
```



The major API functionalities include:

- Dashboard statistics retrieval
- Security alert information
- Traffic analysis data



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- Threat source analysis
- Risk score calculation
- AI model status monitoring
- User profile management
- Report generation
- System configuration management

The APIs act as a communication bridge between the backend processing layer and the SOC dashboard interface.

3. Machine Learning Model Integration

The backend integrates the trained Machine Learning model with the intrusion detection workflow.

The model receives processed network traffic features as input and generates prediction results.

The prediction process includes:

Input Network Features



Machine Learning Model



Traffic Classification



Confidence Score Generation



Severity Assignment



Security Alert Creation

The backend automatically collects the prediction output and stores the results for monitoring and reporting.

4. IDS Log Management

The backend maintains IDS event logs containing detailed information about detected network activities.



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Each log entry includes:

- Event Timestamp
- Source IP Address
- Destination IP Address
- Protocol Information
- Prediction Result
- Confidence Score
- Severity Level
- Alert Status

These logs are continuously processed to update dashboard information and generate security reports.

5. Threat Analysis Module

The backend analyzes detected events to calculate security risk levels and identify potential attack sources.

The threat analysis module performs:

- Attack frequency calculation
- Source IP risk evaluation
- Destination analysis
- Threat score calculation
- Severity classification

This helps security analysts identify high-risk activities and prioritize incident response.

6. Report Generation Module

The backend provides automated report generation functionality for security auditing and documentation.

The reporting module generates:

- CSV Security Reports
- PDF Security Reports

Generated reports include:

- Detected attacks



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- Prediction results
- Confidence values
- Severity information
- Event details

7. Notification Service Integration

The backend includes notification support for critical security events.

The notification module provides:

Email Alerts:

Critical attack notifications can be sent to configured email addresses using SMTP services.

Telegram Alerts:

Instant security alerts can be delivered through Telegram bot integration.

These notification mechanisms help reduce response time during security incidents.

8. User Management and Configuration Module

The backend manages user-related information and system configuration settings.

Functions include:

- User profile management
- Login information tracking
- Severity threshold configuration
- Notification preferences
- Security settings management

Backend Workflow

The complete backend workflow is:

User Request / Network Event



Flask API Server



Data Processing





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Machine Learning Prediction



Threat Analysis



Database / Log Storage



Dashboard Response



Alert Notification

Backend Advantages

The developed backend architecture provides:

- Real-time communication between system components
- Automated threat detection processing
- Efficient security event management
- Scalable API-based design
- Integration capability with future SIEM tools
- Faster incident identification and response

The Flask-based backend architecture forms the foundation of the AI IDS platform by connecting Artificial Intelligence models with practical cybersecurity monitoring and response capabilities.



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4.6 Frontend Modules Explanation

The frontend of the AI Intrusion Detection System (AI IDS) provides an interactive Security Operations Center (SOC) dashboard that enables security analysts to monitor network activities, analyze threats, and manage security incidents efficiently.

The dashboard is designed with multiple modules that provide different security monitoring and management functionalities. Each module presents important security information in a simple and user-friendly format.

AI IDS Login
Secure Analyst Access Portal

Username *
Sahithya

Password *

[Login](#)

[Forgot Password?](#) [Register](#)

LIVE SECURITY INTELLIGENCE

- Initializing SOC feed... ✓ Scanning network traffic...
- ✓ Monitoring IDS engine...
- ✓ Analyzing packet behavior...
- ✓ Firewall status: ACTIVE
- ✓ ML model confidence: 98%
- ✓ No critical threats detected
- ✓ Traffic baseline normal
- ✓ Encrypted packets: 68%
- ✓ System health: STABLE
- ✓ Scanning network traffic...
- ✓ Monitoring IDS engine...
- ✓ Analyzing packet behavior...
- ✓ Firewall status: ACTIVE
- ✓ ML model confidence: 98%
- ✓ No critical threats detected
- ✓ Traffic baseline normal
- ✓ Encrypted packets: 68%
- ✓ System health: STABLE
- ✓ Scanning network traffic...

AI IDS Register
Create your secure analyst account

Full Name *

Email Address *

Phone *

Department *

Role
Security Analyst

Username *

msahithya33@gmail.com

Password *

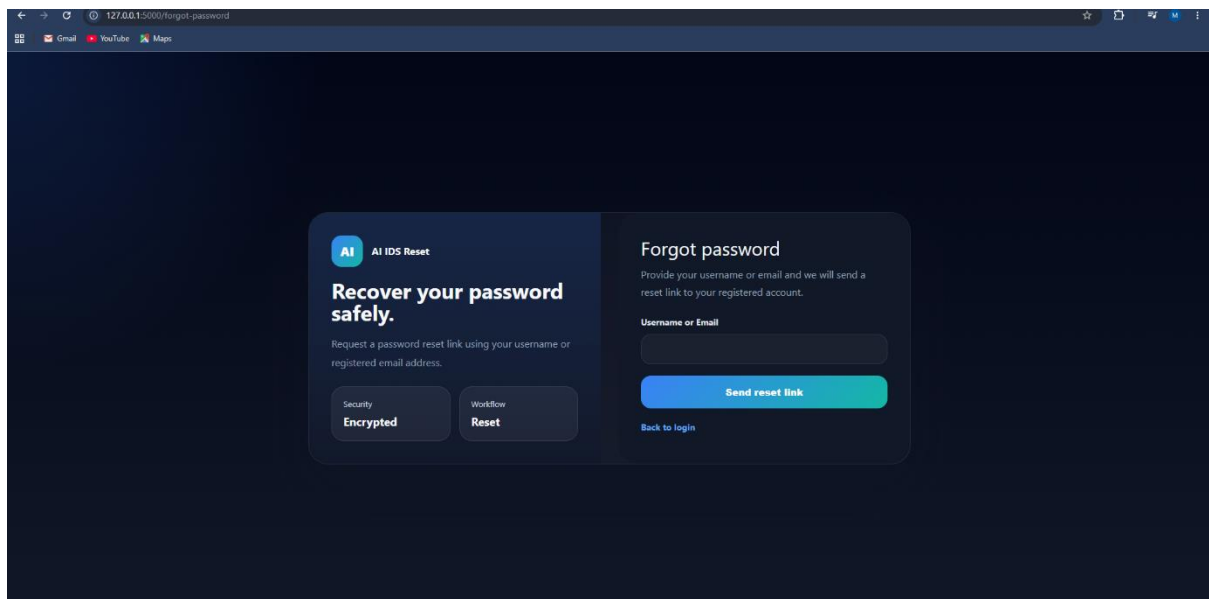
Confirm Password *

[Create Account](#)

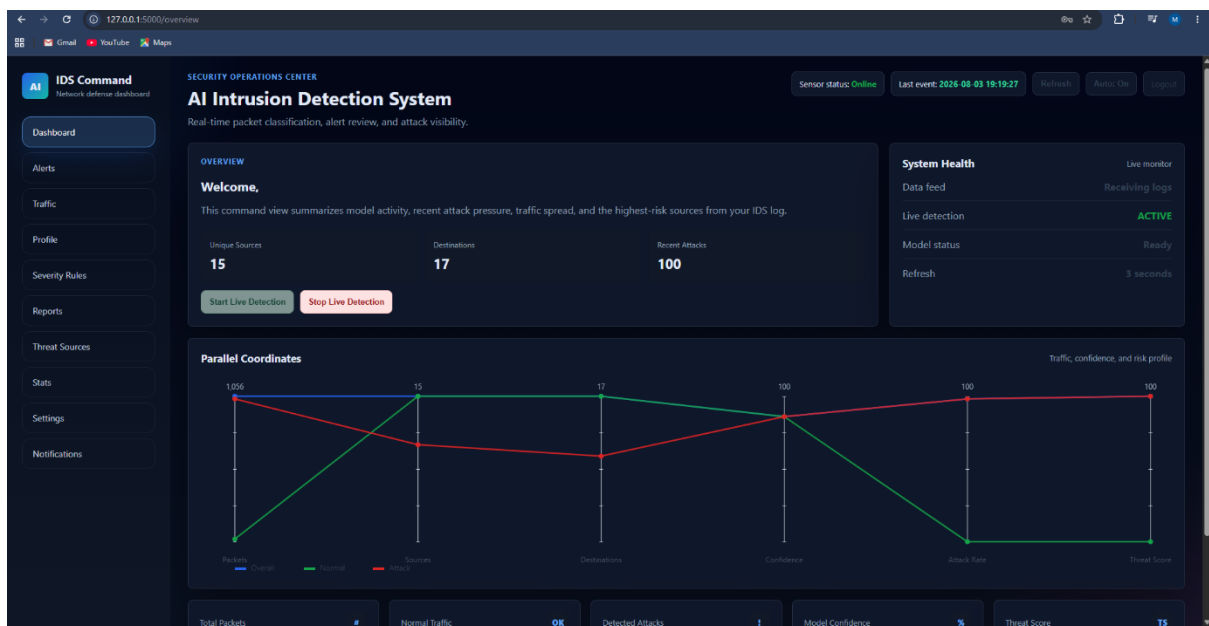
[Already have account? Login](#)



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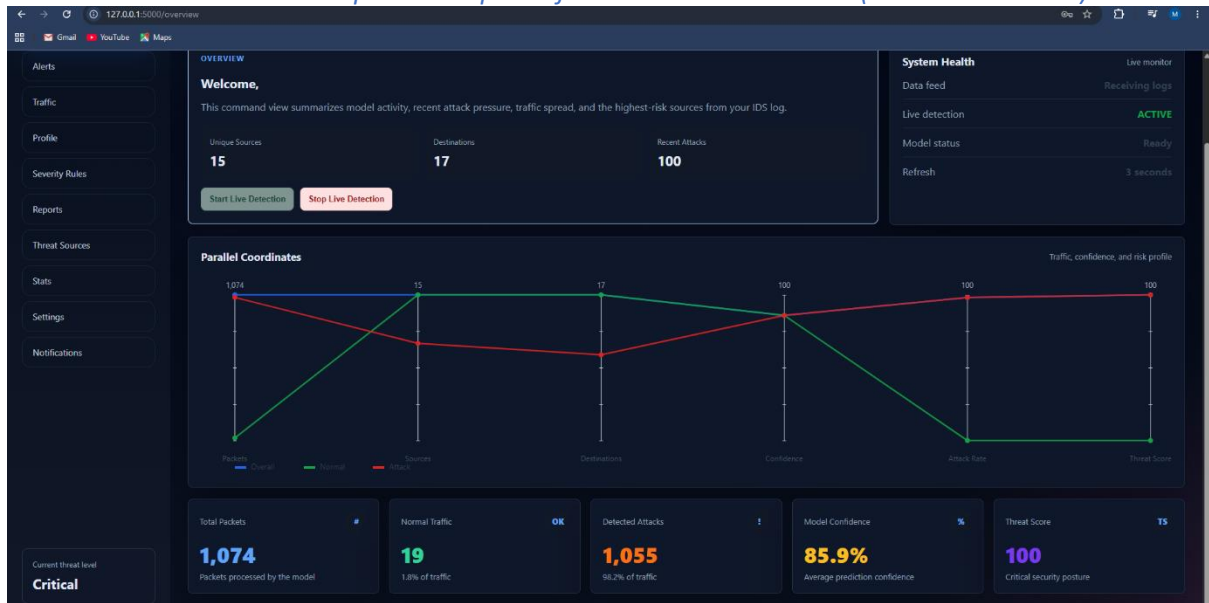


1. Dashboard Module





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The Dashboard module provides an overall summary of the current security status of the network environment.

It displays important security metrics such as:

- Current threat level
- Sensor status
- Model status
- Total packets analyzed
- Normal traffic count
- Detected attack count
- Model confidence score
- Threat score

The dashboard provides real-time visibility into AI model activity and network security conditions. It helps security analysts quickly understand the current threat situation without manually analyzing raw logs.

The module also provides live monitoring functionality with automatic refresh capability to display updated security information.



2. Alerts Module

TIMESTAMP	SOURCE IP	DESTINATION IP	PREDICTION	SEVERITY	CONFIDENCE	STATUS	ASSIGNED	ACTIONS
2025-08-03 19:19:36	35.223.236.179	10.41.109.232	ATTACK	MEDIUM	82%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:34	10.41.109.232	192.178.156.198	ATTACK	MEDIUM	51%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:34	192.178.156.198	10.41.109.232	ATTACK	MEDIUM	76%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:34	10.41.109.232	35.223.236.179	ATTACK	MEDIUM	76%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:32	35.223.236.179	10.41.109.232	ATTACK	MEDIUM	82%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:32	10.41.109.232	34.49.39.67	ATTACK	CRITICAL	96%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:32	34.49.39.67	10.41.109.232	ATTACK	HIGH	91%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:27	10.41.109.242	10.41.109.232	ATTACK	MEDIUM	94%	OPEN	Analyst	Open Ack Save
2025-08-03 19:19:27	10.41.109.242	10.41.109.232	ATTACK	MEDIUM	94%	OPEN	Analyst	Open Ack Save

The Alerts module is responsible for displaying detected security incidents generated by the AI IDS model.

It provides detailed information about each detected event, including:

- Timestamp
- Source IP Address
- Destination IP Address
- Prediction Result
- Severity Level
- Confidence Score
- Alert Status
- Assigned Analyst

The module supports features such as:

- Searching alerts using IP addresses or prediction results
- Filtering alerts based on severity
- Viewing attack and normal traffic events

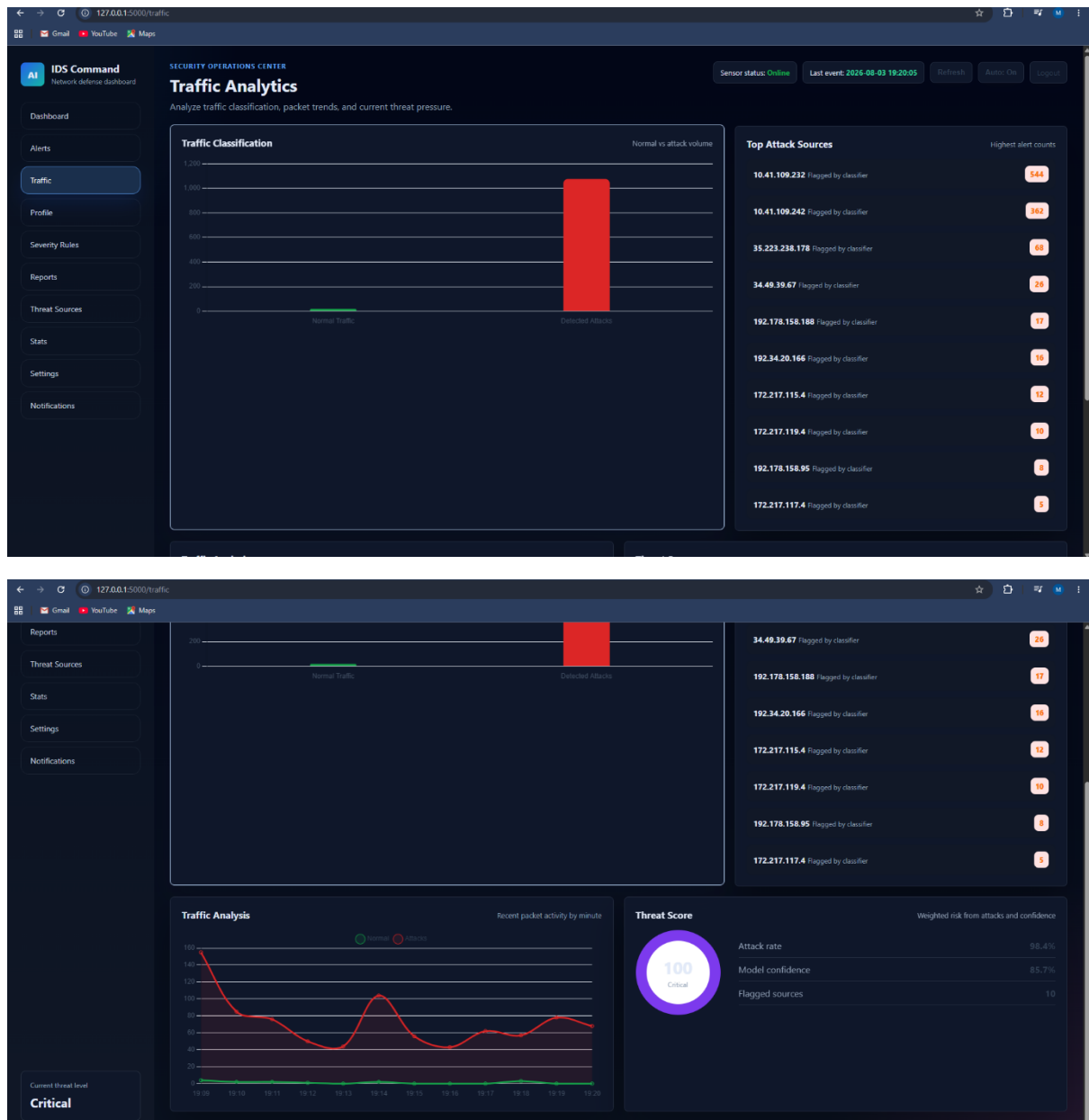


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- Exporting security alert data

This module helps security analysts investigate suspicious activities and prioritize important incidents.

3. Traffic Analytics Module



The Traffic Analytics module provides graphical analysis of network traffic behavior.

It displays:

- Normal traffic volume
- Attack traffic volume



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- Traffic classification results
- Attack rate percentage
- Model confidence
- Flagged source information
- Threat score analysis

The visualization helps analysts understand traffic patterns and identify abnormal network behaviour.

This module improves security visibility by converting raw traffic data into meaningful security insights.

4. Profile Module

The screenshot displays the 'User Profile' page within the 'IDS Command' interface. The page is titled 'SECURITY OPERATIONS CENTER User Profile' and provides a view of the account, access, and database information for the signed-in user, 'Sahithya', who is a Security Analyst. The interface includes a sidebar with navigation options such as Dashboard, Alerts, Traffic, Profile, Severity Rules, Reports, Threat Sources, Stats, Settings, and Notifications. The main content area is divided into several sections: a 'SIGNED-IN USER' section showing the user's name, role, and department; a 'Complete User Information' table with fields like Full Name, Username, Email Address, Phone, Role, Department, Email Verified, Account Created, Last Login, Login Count, User Storage, and Account record; and an 'Edit Profile' section at the bottom with input fields for Full Name, Email Address, Phone, and Department. The page also features a 'Sensor status: Online' indicator and a 'Last event: 2026-08-03 19:20:24' timestamp.

Complete User Information		Account record	
Full Name	Sahithya	Username	Sahithya
Email Address	msahithya33@gmail.com	Phone	Not provided
Role	Security Analyst	Department	Security
Email Verified	No	Account Created	Not recorded
Last Login	2026-08-03 19:19:02	Login Count	37
User Storage	users.json		



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The screenshot displays a user profile page for 'Sahithya', a Security Analyst. The page is divided into several sections:

- SIGNED-IN USER:** Sahithya, Security Analyst.
- User Details:** Username: Sahithya, Department: Security, Database: users.json.
- Account Information:** Email Address: msahithya33@gmail.com, Phone: Not provided, Role: Security Analyst, Last Login: 2026-08-03 19:19:02, Login Count: 37, User Storage: users.json.
- Edit Profile:** Fields for Full Name (Sahithya), Email Address (msahithya33@gmail.com), Phone (Not provided), Department (Security), Role (Security Analyst), and a checkbox for 'Notify on critical alert' (checked).
- Current threat level:** Critical.

The Profile module manages security analyst account information and user details.

It provides:

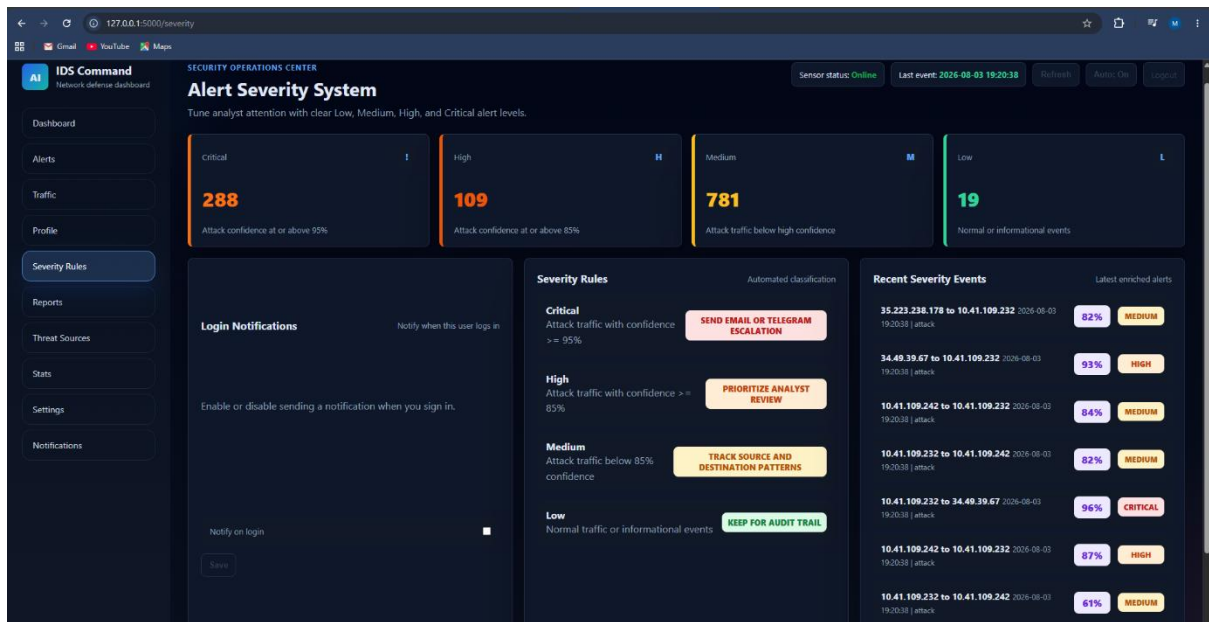
- Username information
- User role
- Department details
- Email information
- Account information
- Login history

The module also allows users to update profile details and configure notification preferences.

This improves user management and provides personalized access information within the SOC dashboard.



5. Severity Rules Module



The Severity Rules module provides automated classification of security alerts based on threat confidence levels.

The system categorizes alerts into:

Critical:

Attack confidence equal to or above 95%.

High:

Attack confidence equal to or above 85%.

Medium:

Attack traffic with moderate confidence values.

Low:

Normal or informational security events.

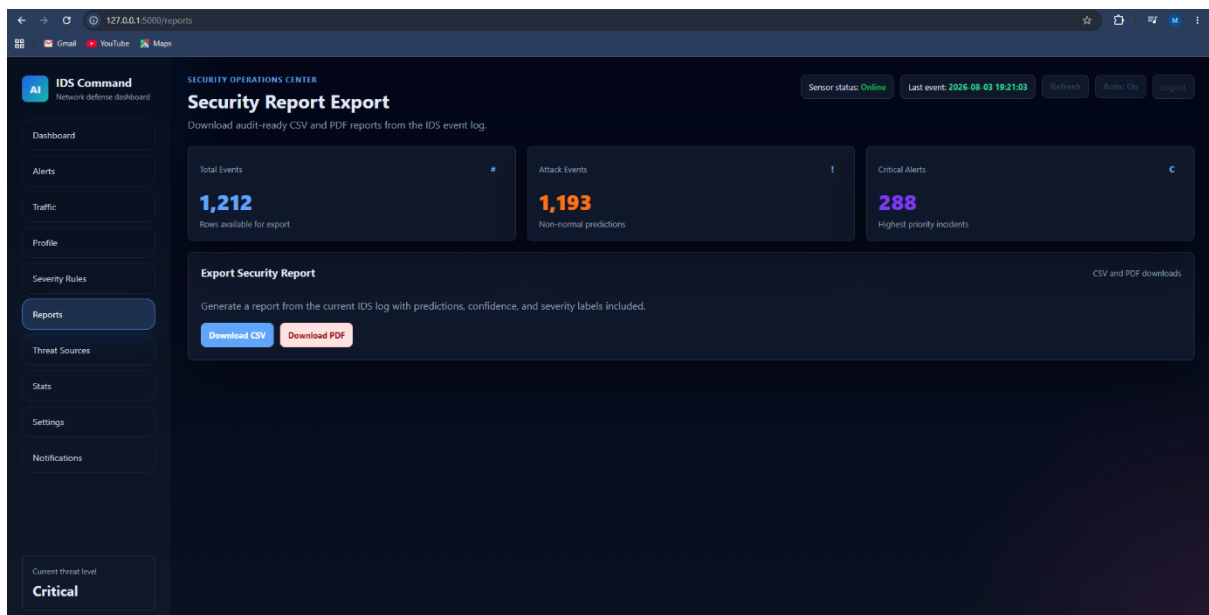
This classification helps analysts prioritize critical threats and focus on high-risk incidents first.

The module also allows customization of severity thresholds according to organizational security requirements.



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6. Reports Module



The Reports module provides security report generation functionality for auditing and documentation purposes.

The system supports:

- CSV report generation
- PDF report generation

The generated reports contain:

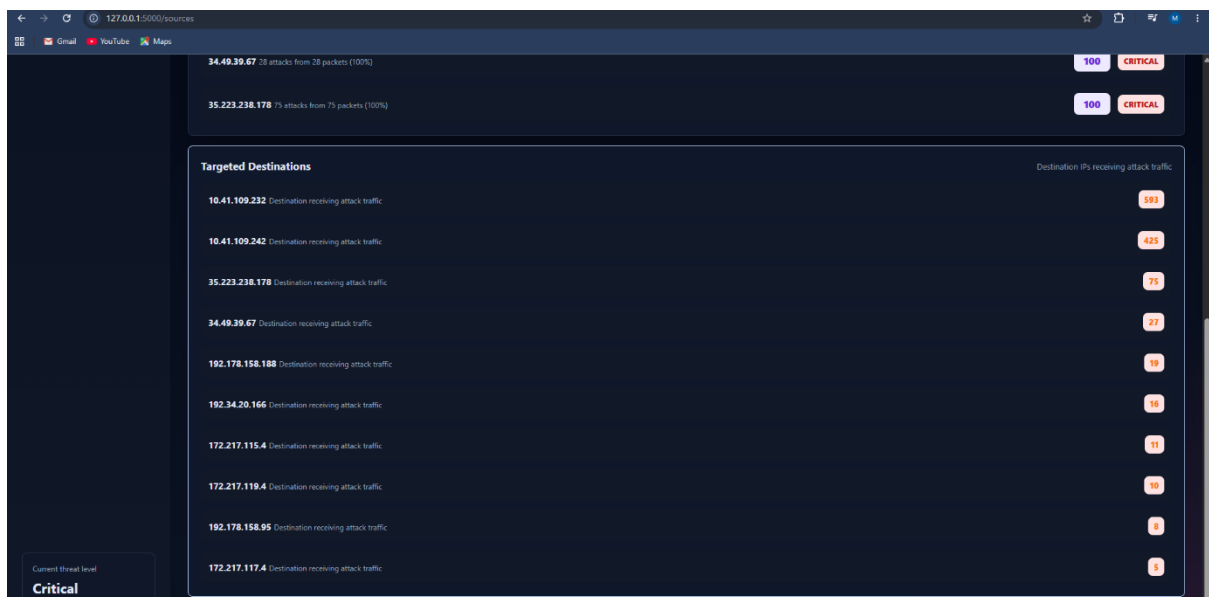
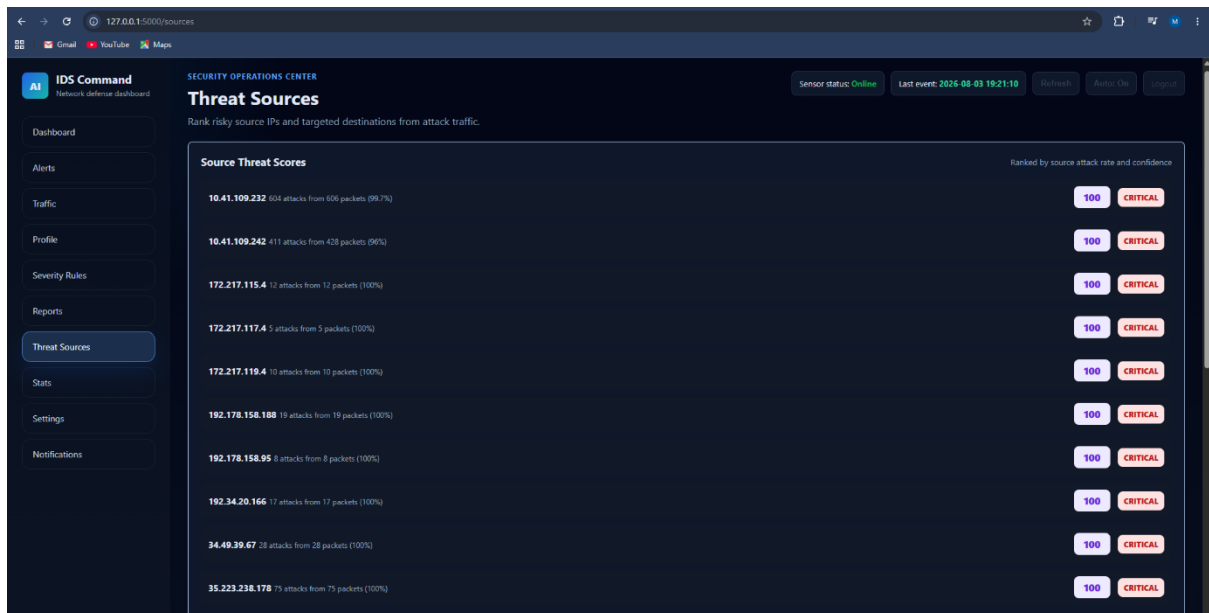
- Security event details
- Prediction results
- Confidence scores
- Severity classifications
- Attack information

This module helps organizations maintain security records and perform incident analysis.



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7. Threat Sources Module



The Threat Sources module identifies risky network sources involved in malicious activities.

It provides analysis of:

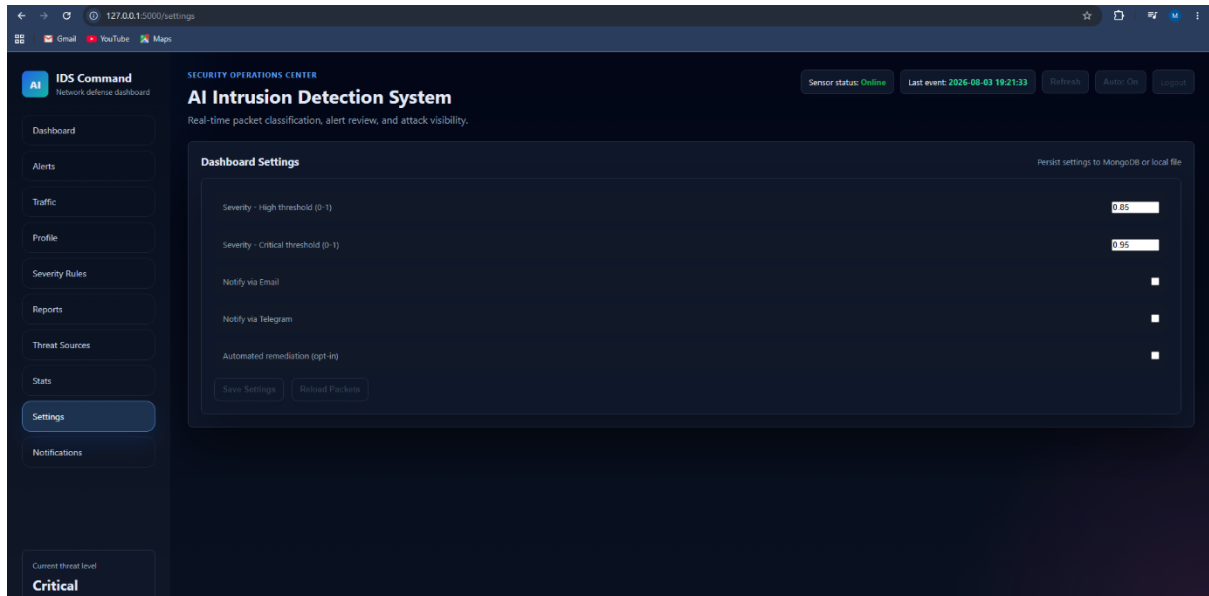
- High-risk source IP addresses
- Attack frequency
- Source threat scores
- Targeted destination systems

This module helps security analysts identify the origin of attacks and understand possible attack patterns.



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8. Settings Module



The Settings module provides configuration options for customizing the AI IDS system.

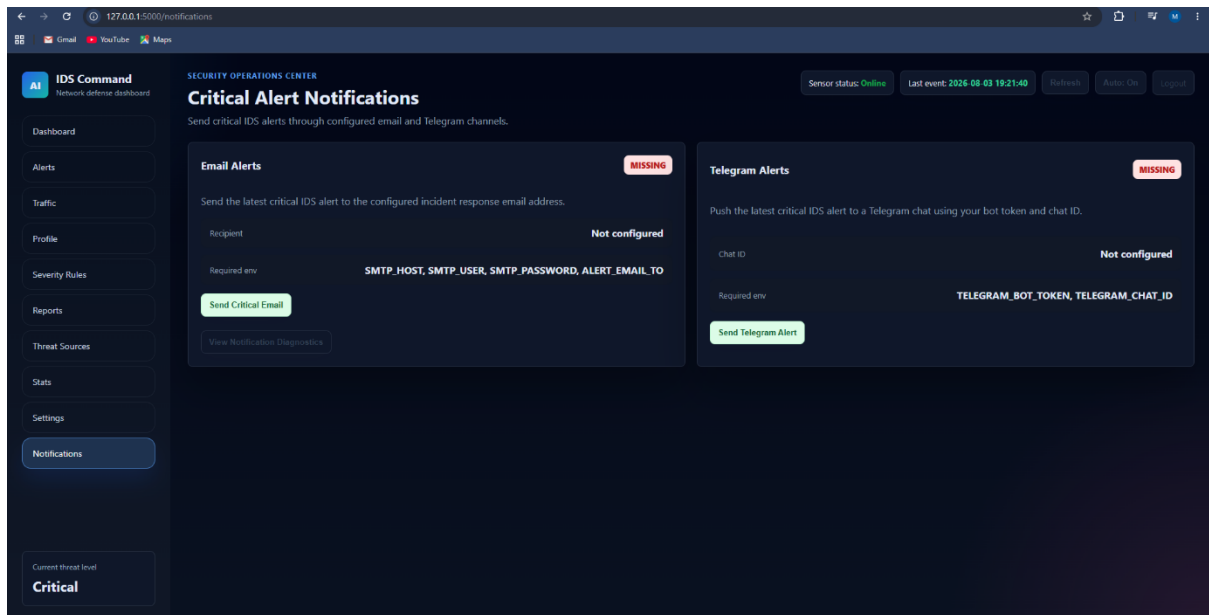
Available settings include:

- High severity threshold configuration
- Critical severity threshold configuration
- Email notification settings
- Telegram notification settings
- Automated remediation options

These configurations allow administrators to modify system behavior based on security requirements.



9. Notifications Module



The Notifications module provides automated alert delivery mechanisms for critical security incidents.

The system supports:

Email Notifications:

Critical IDS alerts can be sent through configured email channels.

Telegram Notifications:

Security alerts can be delivered instantly through Telegram bot integration.

This feature enables faster incident response and reduces the time required to notify security teams.

Frontend Module Summary

The AI IDS frontend dashboard provides a centralized security monitoring platform by combining real-time visualization, alert management, threat analysis, reporting, and notification capabilities.

Through these modules, the system enables security analysts to monitor network activities, detect threats, analyze attacks, and respond effectively to cybersecurity incidents.



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4.8 GitHub Repository Link

The complete source code of the AI Intrusion Detection System (AI IDS) project is maintained using GitHub for version control, collaboration, and future development.

The repository contains all major components of the project, including:

- Backend application source code
- Machine Learning model implementation
- Data preprocessing modules
- IDS log processing components
- Frontend dashboard code
- API integration files
- Configuration files
- Project documentation

GitHub provides a centralized platform for managing project updates, tracking changes, and maintaining different versions of the application.

GitHub Repository:

<https://github.com/sure-trust/MITTA-VENKATA-NAGA-SAHITHYA-g15-cs/tree/main>



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The repository demonstrates the complete implementation of an AI-powered cybersecurity monitoring platform and allows developers and security researchers to understand, extend, and improve the system in the future.

Social/Industry Relevance

Cybersecurity has become one of the most critical challenges for organizations due to the continuous increase in cyber attacks, data breaches, and network vulnerabilities. Traditional security monitoring methods often depend on predefined rules and signatures, making them less effective against new and evolving threats.

The AI Intrusion Detection System addresses this challenge by applying Artificial Intelligence and Machine Learning techniques to automatically analyze network traffic and identify suspicious activities.

Industry Relevance:

The developed system has practical applications in various cybersecurity environments, including:

- Security Operations Centers (SOC)
- Enterprise Network Monitoring
- Cloud Security Platforms
- Banking and Financial Security Systems
- Government Infrastructure Protection
- Educational and Research Networks

The project provides security analysts with:



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- Automated threat detection
- Real-time security monitoring
- Intelligent alert classification
- Attack source identification
- Security report generation
- Faster incident response

Social Impact:

The AI IDS platform contributes to improving digital security awareness and protecting sensitive information from cyber threats. By reducing manual monitoring efforts, the system helps security teams respond quickly to potential attacks.

The project also demonstrates how Artificial Intelligence can be used as a practical cybersecurity solution to build safer and more reliable digital environments.



Learning and Reflection

Developing the AI Intrusion Detection System provided valuable practical experience in the fields of Artificial Intelligence, Machine Learning, Cybersecurity, Backend Development, and Security Operations.

During the project implementation, several technical and professional skills were gained.

Technical Learning:

- Understanding of Intrusion Detection Systems and cybersecurity concepts.
- Practical implementation of Machine Learning algorithms for attack classification.
- Experience in preprocessing and analyzing network traffic data.
- Development of REST APIs using Flask framework.
- Integration of Machine Learning models with web applications.
- Designing and developing interactive SOC dashboards.
- Working with security logs and threat analysis techniques.
- Implementing automated security reporting and notification mechanisms.

Problem-Solving Experience:

During the development process, challenges related to data preprocessing, model integration, API communication, and dashboard development were addressed through debugging, testing, and continuous improvement.

The project improved skills in:

- Logical thinking
- Debugging techniques
- System design
- Research and analysis
- Documentation practices

Professional Learning:

This project provided practical exposure to how real-world Security Operations Centers monitor, analyze, and respond to cybersecurity incidents. It helped in understanding the importance of automation, threat intelligence, and Artificial Intelligence in modern cybersecurity.

Overall, the project enhanced technical knowledge and provided hands-on experience in developing an end-to-end AI-powered cybersecurity solution.



Conclusion and Future Scope

Future Scope

Although the developed AI Intrusion Detection System provides effective network monitoring and threat detection capabilities, several improvements can be implemented in future versions to enhance accuracy, scalability, and automation.

The possible future enhancements are:

1. Real-Time Network Traffic Capture

The current system can be extended by integrating real-time packet capture tools such as Wireshark, Scapy, or network sensors. This will allow the system to directly monitor live network traffic and detect threats immediately.

2. Deep Learning-Based Threat Detection

Future versions can incorporate advanced Deep Learning techniques such as:

- Artificial Neural Networks (ANN)
- Convolutional Neural Networks (CNN)
- Long Short-Term Memory Networks (LSTM)
- Transformer-based models

These approaches can improve detection accuracy, especially for complex and previously unknown attack patterns.

3. Integration with SIEM Platforms

The AI IDS can be integrated with Security Information and Event Management (SIEM) platforms such as:

- Wazuh
- Splunk
- Microsoft Sentinel
- Elastic Security

This integration can provide centralized security monitoring and advanced threat intelligence capabilities.

4. Automated Incident Response

Future development can include automated response mechanisms such as:



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- Blocking malicious IP addresses
- Isolating compromised systems
- Updating firewall rules
- Executing predefined response actions

This will reduce manual intervention and improve incident response time.

5. Cloud-Based Deployment

The system can be deployed on cloud platforms to support scalable security monitoring for large organizations. Cloud deployment enables:

- Distributed monitoring
- High availability
- Large-scale data processing
- Remote security management

6. Threat Intelligence Integration

Future versions can integrate external threat intelligence sources to identify newly emerging cyber threats and improve attack prediction capabilities.

7. Explainable Artificial Intelligence (XAI)

Explainable AI techniques can be implemented to provide reasons behind Machine Learning predictions. This helps security analysts understand why a particular network activity was classified as malicious.

8. Advanced User Management and Access Control

The system can be enhanced with:

- Role-based access control
- Multiple analyst accounts
- Authentication improvements
- Security audit tracking

9. Mobile Security Monitoring Application

A mobile application can be developed to allow security analysts to monitor alerts and receive critical notifications remotely.

The future enhancements will transform the AI IDS platform into a complete autonomous cybersecurity solution capable of detecting, analyzing, and responding to advanced cyber threats.



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8. Conclusion

The AI Intrusion Detection System (AI IDS) successfully demonstrates the application of Artificial Intelligence and Machine Learning techniques in improving modern cybersecurity operations.

The main objective of this project was to develop an intelligent system capable of monitoring network traffic, detecting malicious activities, generating security alerts, and providing security analysts with a centralized monitoring platform.

The developed system combines multiple technologies including Machine Learning, Flask backend services, REST APIs, data processing techniques, and an interactive SOC dashboard to provide an efficient intrusion detection solution.

The Machine Learning model analyzes network traffic patterns and classifies activities into normal and attack categories. Based on the prediction results and confidence scores, the system automatically generates alerts and assigns appropriate severity levels.

The SOC dashboard provides complete visibility into security events through different modules including:

- Dashboard monitoring
- Live security alerts
- Traffic analytics
- Threat source analysis
- Severity management
- Security report generation
- Notification services

The project successfully achieves automated threat detection, real-time security visualization, and improved incident monitoring capabilities.

Through this project, practical knowledge was gained in cybersecurity concepts, Machine Learning implementation, backend development, API integration, and security operations.

In conclusion, the AI Intrusion Detection System provides an effective foundation for intelligent cybersecurity monitoring and can be further enhanced with advanced AI models, real-time network integration, SIEM connectivity, and automated incident response mechanisms to meet future cybersecurity challenges.



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